On the applicability of conventional stability-based paradigms of ocean interior turbulent mixing

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Key Points:

- Conventional marginal-stability parameterizations of ocean mixing seem to miss the continuum of mixing in ocean interiors
- The deficiency is linked to spatial biases and lack of representation of the diversity of mixing processes
- Observations combined with novel data-driven methods (simulations, data assimilation, machine learning) can help identify the knowledge gaps

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Abstract

Understanding how turbulence enhances irreversible scalar mixing in stratified fluids is a central problem in geophysical fluid dynamics, with the variability of such turbulent processes affecting ocean circulation and budgets of tracers such as nutrients and carbon. However, measuring mixing rates in the ocean and resolving mixing processes in climate models is currently out of reach because of the (very) small length scales involved. Hence, most observations and models rely upon parameterisations of mixing mechanisms that inevitably are based on assumptions of the characteristics of the underlying turbulence. Here, using realistic simulations of the Drake Passage, we analyse potential biases introduced by commonly used assumptions. More precisely, we show that marginal instability-based parameterisations might be biased by undersampling when applied at scales relevant to, e.g. climate models. In particular, we show that while the abyssal ocean might be well described by a marginal instability paradigm of mixing, such a description is inaccurate in the upper- and mid-ocean when applied to relatively coarse resolution models and observations and might miss fundamental mixing processes. We thus question the applicability of the marginal stability-based parameterisations of deep ocean mixing widely used in the ocean component of Earth System Models.

Plain Language Summary

Turbulence enhances the rate at which waters with different salt and heat contents mix. This phenomenon plays a crucial role in setting the large-scale global ocean circulation, controlling the ocean's biological productivity and guiding the pathway of climatically relevant tracers such as carbon. However, mixing by turbulent motion happens on extremely small scales that are not yet resolved by ocean and climate models. Therefore, mixing by turbulent motions needs to be inferred from indirect measurements and then be parameterised for climate models. This work aims to analyse the potential biases arising from commonly used parameterisations of turbulent mixing. Our findings suggest that whereas such parameterisations might do better in the abyssal ocean, they are not accurately resolving turbulent mixing in the upper- and mid-part of the water column.

1 Introduction

Turbulence is known to enhance the rate at which fluid parcels with different scalar properties (for example, heat, salinity and nutrients) irreversibly mix within the ocean. Mixing is a process that occurs on relatively small scales (of order 0.01-1 mm or smaller) and yet is known to have a leading order impact on the large-scale global ocean circulation (Wunsch & Ferrari, 2004), biological productivity (Marra et al., 1990), and tracer pathways (Cimoli et al., 2023). Quantifying the rate at which turbulence mixes scalar gradients and hence restratifies the water column is therefore of crucial importance in a variety of geophysical and environmental settings. Yet, due to the small scales involved, it is currently impossible to resolve mixing directly in observational measurements and large-scale models of the oceans, and so approximate mixing proxies and parameterisations are required.

Microstructure measurements of the turbulent dissipation rates of kinetic energy ϵ and thermal variance χ currently provide the best observational proxies for approximating mixing rates from available oceanographic measurements (Gregg et al., 2018). These quantities describe the rates at which viscosity and molecular diffusion smooth velocity and scalar gradients, respectively, which can then be used to infer scalar turbulent diffusivities that are required inputs to mixing parameterisations in large-scale ocean models. For instance, turbulent diffusivities of density are inferred from χ using the Osborn-Cox method (Osborn & Cox, 1972) and from ϵ using the Osborn method (Osborn, 1980).

Other methods that do not directly rely on microstructure measurements, but instead infer a measure of mixing from more rudimentary scalar and velocity field measurements,

also exist. We focus on these methods in this work. One such method relies on the concept of self-organized criticality (SOC). Using *in situ* measurements, Smyth et al. (2019) provided evidence that events with relatively elevated values of ϵ are characterised by a Richardson number Ri (quantifying the relative strength of stratification compared to background vertical shear in the flow) close to $Ri_c \simeq 1/4$, the critical value below which it is possible for a laminar, steady inviscid stratified shear flow to become unstable (Miles, 1961; Howard, 1961). As a rule of thumb, for an initially laminar flow forced such that the shear continuously increases, its associated Richardson number will eventually drop below the critical value, hence becoming unstable and breaking up into turbulence. The generation of turbulence then diffuses momentum, allowing the Richardson number to grow until the flow becomes stable again, after which a new instability cycle can begin. Such a flow may then be thought of residing in a continual state of marginal stability, provided that it is assumed that the turbulence is entirely driven by shear instabilities. Assuming such marginal organisation of the flow potentially enables the construction of a marginal 'mixing efficiency' around which the flow should self-organise (Smyth, 2020) and suggests that mixing in the ocean interior could be parameterised in terms of the Richardson number only.

This method is supported by a fundamental assumption, namely that mixing is inherently related to particular instabilities (shear-instabilities) that mix the water column at just the right rate so that these instabilities remain in a marginally unstable state. It is important to question this assumption and its consequences. More precisely

1. The marginal instability paradigm of Smyth et al. (2019) has been observed in relatively high-resolution regional models of the ocean (see for instance figure 1 of (Mashayek, Baker, et al., 2022)). These models usually rely on mixing parameterisations based on the marginal instability ideas presented above. It is therefore natural to ask whether such findings are a consequence of the parameterisation used in the models rather than of physical properties of the ocean.
2. It is important to note that the above considerations about self-organised criticality were observed when all (or almost all) scales were available. For instance, Smyth et al. (2019) considered profiles resolving density inversions with vertical extent as small as 2 cm to deduce their SOC results. This raises a fundamental question: do these results survive downscaling? Are they still relevant when applied at the relatively large scale of interest to, e.g., climate models, i.e. when no measurement at the small scales is available? Practically, is a Richardson number-based parameterisation accurate when applied at the resolution scale of a relatively large-scale ocean model, i.e. when the Richardson number can only be estimated on a relatively coarse grid?
3. Finally, are parameterisations of mixing based on the marginal instabilities ideas presented above biasing statistics towards particular regions of the water column, where such instabilities are more likely to happen? Are they overlooking other processes that might be as important as shear instabilities in setting turbulent statistics? For instance, Smyth et al. (2019) have suggested that Ri is log-normally distributed. Then, the assumption of marginal stability neglects the large-Ri tails commonly observed in the Ri probability distribution, corresponding to flows stable to shear instability but which have the potential also to contribute significantly to mixing.

To answer these questions, we consider a common Richardson number-based parameterisation of turbulent diffusivity applied to a realistic simulation of the Drake Passage in the Southern Ocean and compare its output with an alternative data-driven parameterisation that is trained using observational data. We demonstrate that, when applied at the resolution scale of the simulations considered here, the Richardson number-based predictions are inconsistent with observationally constrained estimates of turbulent diffusivities. More precisely, the Richardson number-based parameterisation restratifies the water column at just the right rate so that the flow is statistically in a marginally unstable state with $Ri \simeq Ri_c$. Such a result is not observed when considering the observationally constrained parameterisation of mixing. Furthermore, our findings suggest that current parameterisa-

tions of mixing and large-scale ocean models are potentially biased towards the bottom of the water column, at least when applied on scales relevant to climate models. This raises the following question: are these inherently small-scale-based parameterisations adapted to coarse resolution modelling?

2 Biases in marginal instability-based approaches

Here, we analyse the potential biases arising from parameterisations based on the Richardson number. The Richardson number is defined as $Ri := N^2/S^2$, where the buoyancy frequency $N^2 := -\frac{g}{\rho_0}\partial_z\rho$ is proportional to the vertical gradient of the density ρ (g is the gravitational acceleration and ρ_0 is a reference density) and $S^2 = (\partial_z u)^2 + (\partial_z v)^2$ is the squared vertical shear (where u, v are horizontal velocities). Here, ρ refers to the neutral density (Jackett & McDougall, 1997). Neutral density is a variable that eliminates dynamically irrelevant density variations due to compressibility effects and that is not assigned to a reference pressure point (Jackett & McDougall, 1997) and will be henceforth referred to as density and denoted ρ for simplicity.

The Richardson number Ri is a measure of the relative strengths of the background stratification and shear in a stratified-shear flow. As discussed in the Introduction, microstructure oceanic measurements show that turbulent events with relatively elevated dissipation rates ϵ and associated (inferred) turbulent scalar diffusivity κ_T are characterised by values of Ri close to the Miles-Howard critical stability value $Ri_\mathrm{c} = 1/4$ (Miles, 1961; Howard, 1961), suggesting that such events are marginally unstable if we assume that turbulence in the oceans is entirely driven by shear instabilities (Smyth et al., 2019). This apparent self-organisation of the flow suggests that parameterisations of ϵ and κ_T based on Ri only should provide an accurate picture of the overall statistics of dissipation rates and diffusivity in the water column, especially for the extreme events which often dominate mixing statistics. We would like to test this hypothesis, especially when only limited information about the underlying microstructure of the flow is available.

2.1 Method

To do so, we consider in this work the commonly used the K-Profile Parameterisation (KPP) (Large et al., 1994). This parameterisation (outside of the ocean surface boundary layer) is exclusively based on a Richardson number criterion as described above: the lower the Richardson number, the more likely a shear-instability will appear, thus overturning density gradients, breaking down into turbulence, dissipating kinetic energy and momentum and finally enabling the Richardson number to grow again so that a marginal state is restored. This approach also enables the parameterisation of convective instabilities, where $N^2 < 0$ and correspondingly $Ri < 0$. Hence, it is assumed that it suffices to parameterise the dissipation of energy and momentum using Ri alone, and this is exactly what KPP is designed for; given a Richardson number, KPP outputs a proxy of how shear-driven turbulence dissipates momentum. This proxy is commonly referred to as the 'turbulent diffusivity of momentum' (denoted ν_T) and is related to a proxy for mixing — the 'turbulent scalar diffusivity' κ_T that describes how turbulence diffuses the scalar being mixed and hence smooths density gradients — by assuming $\nu_\mathrm{T} \simeq \kappa_\mathrm{T}$ (or equivalently, a turbulent Prandtl number $Pr_\mathrm{T} := \nu_\mathrm{T}/\kappa_\mathrm{T} \simeq 1$).

We would like to compare the above parameterisations with actual *in situ* measurements. Note that the turbulent diffusivity κ_T mentioned above cannot be measured directly in the oceans (it is an abstract construction designed for modelling and operational purposes) and is defined via the Osborn relation (Osborn, 1980) as:

$$\kappa_\mathrm{T} := \Gamma \frac{\epsilon}{N^2}, \tag{1}$$

where Γ is the 'flux coefficient' and ϵ is the dissipation rate of turbulent kinetic energy, a quantity that can be measured. Hence, using microstructure *in situ* data, one can retrieve

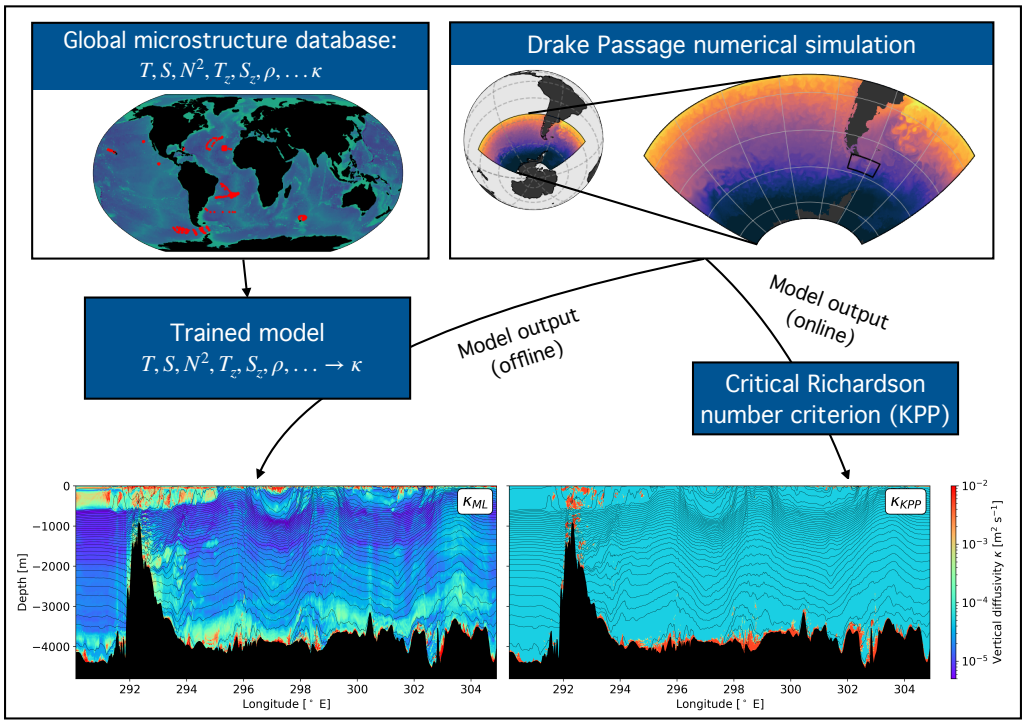


Figure 1. Project framework: an observationally-tuned numerical simulation of the Drake Passage in the Southern Ocean (upper right) is used to enable comparison between observed and parameterised turbulent vertical diffusivities κ_{T} in the ocean. A global microstructure database (upper left; red dots correspond to the geographical locations where the microstructure data were collected) is used to train a machine learning (ML) model (Mashayek, Reynard, et al., 2022) to predict turbulent diffusivities κ_{T} given features that are available in both the observational dataset and simulation, which is then applied to the data arising from the simulation. An example of the resulting diffusivity κ_{ML} inferred 'offline' using the ML model is shown in the lower left panel. The diffusivity κ_{KPP} calculated using the KPP scheme 'online' within the evolving simulation is presented in the lower right panel. Note that, by definition of the KPP parameterisation, the red region has $Ri < Ri_c$ and the light blue one (with diffusivities equal to the background value 5×10^{-5} m^2 s^{-1}) has $Ri > Ri_c$.

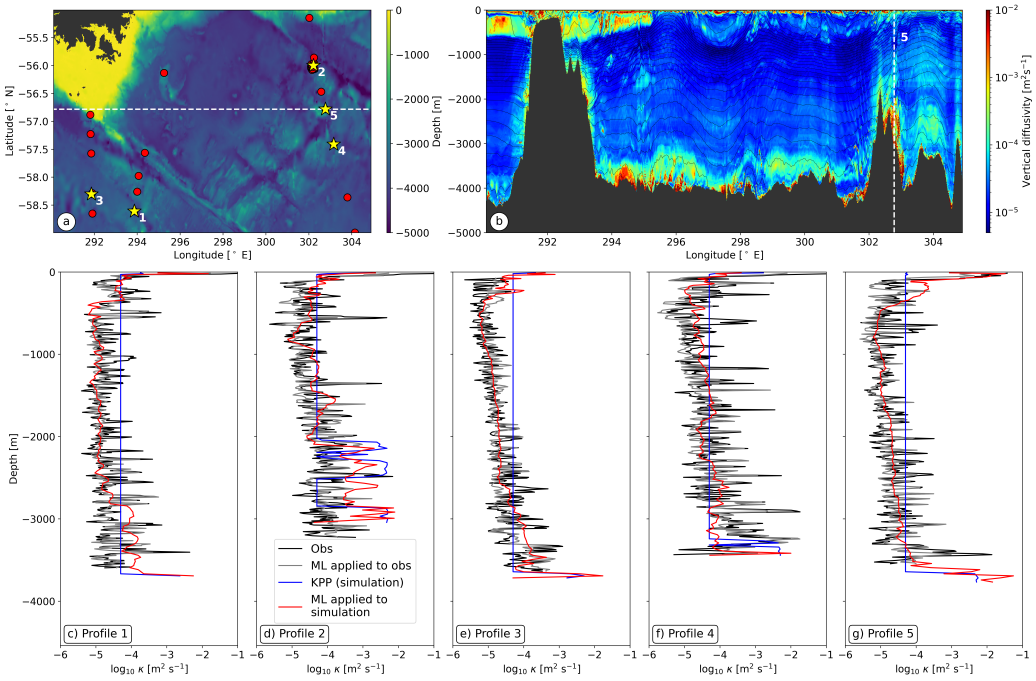


Figure 2. (a) The domain of the Drake Passage simulation, with colour showing depth. Coloured symbols denote locations of microstructure profiles, with yellow stars indicating the test profiles shown in the bottom row, and red circles indicating training data in the domain, which are a subset of the global microstructure data used to train the model (locations of which are shown in figure 1). The white dashed line corresponds to the section shown in panel (b). (b) A section of diffusivity calculated by applying the ML model to the simulation output. The vertical dashed white line corresponds to profile 5 in the bottom row. (c-g) Diffusivity profiles at the locations shown in panel (a). Observed diffusivities are shown in black, and the corresponding ML model output (trained on all data except for these 5 profiles) is shown in grey. The ML model output when applied to the simulation output is shown in red; note that we do not expect exact correspondence between the red and black/grey lines due to temporal variability in diffusivities. The 'online' diffusivity κ_{KPP} calculated in the simulation using the KPP scheme is shown in blue.

an estimate of how turbulence diffuses a scalar through κ_{T} (for the purpose of our work, we *assume* $\Gamma = 0.2$, a commonly used assumption in operational physical oceanography (Gregg et al., 2018)). We would like to compare these real-world estimates with the KPP parameterisation presented above. To do so, we would ideally need collocated measurements of ϵ (and thus κ_{T} through equation 1) and Ri. However, datasets containing both microstructure and Ri measurements are rare, making it difficult to assess the robustness of a Richardson number-based approach to mixing. In this work, we therefore develop a method to bring together oceanic microstructure observations and process-resolving and realistic simulations of the Drake Passage (a well known hot-spot of turbulent mixing (Garabato et al., 2004; Sheen et al., 2013; Mackay et al., 2018)) in order to build a synthetic (but importantly, observationally constrained) dataset that can be used as a test case for the KPP parameterisation. The method can be described as follows, and is summarised in figure 1.

First, following the methodology of Mashayek, Reynard, et al. (2022) (hereafter M22), we train a supervised machine learning model on existing observational measurements of

temperature, salinity, their vertical gradients, (the logarithm of) buoyancy stratification, latitude, depth, and height above bottom, to learn (the logarithm of) turbulent diffusivity κ_T, which has itself been 'directly' calculated from microstructure measurements of ϵ via equation 1. Whilst M22 compared a neural network and a bootstrap aggregated decision tree approach for this problem, we apply an XGBoost regressor model, for its simple implementation and known high performance when predicting tabular data such as that considered here (Chen & Guestrin, 2016). In addition to the global microstructure dataset containing $\sim 200,000$ data points distributed over 702 profiles (locations are shown in figure 1), we also include a further set of training data consisting of 46 full-depth profiles ($\sim 20,000$ data points) taken by the DIMES observational campaign in the Drake Passage (Sheen et al., 2013) in order to improve predictive skill in this region of the ocean.

The models constructed by M22 were validated by 10-fold cross validation over the entire global microstructure dataset, with the train-test split constructed by random sampling. By this metric, our algorithm performs similarly to the neural network in M22, giving an R^2 score on 10-fold cross-validation of 0.72 for predicting $\log \kappa_T$. However, by this method of validation, neighbouring data points in a single profile are treated independently, insofar as one may end up in the training dataset and the other may end up in the test dataset. This gives the model a significant predictive advantage. Here, we wish to apply the ML model to unseen simulation data in the Drake Passage, and we therefore validate our model on Drake Passage profiles that are completely absent from the training data.

Figure 2(a) shows the Drake Passage simulation domain, with the 33 available observational profile locations in the domain shown by red and yellow markers. To test the ability of our model on these profiles, we perform an 11-fold cross-validation; we randomly sample 3 profiles and withhold these as test data. We then train on the remaining profiles and all other global data, and calculate the R^2 score when predicting $\log \kappa_T$ on the test data. This is repeated 10 more times over the 33 profiles without replacement, so that the model is tested on every Drake Passage profile. The entire process is repeated 10 times to reduce variance, and the resulting mean R^2 score is 0.42. For the reasons discussed above, this is less than the R^2 score of the model when the test data is randomly chosen over all data points, but shows that the model is still able to achieve reasonable performance on unseen profiles in the Drake Passage. Examples of this performance are shown in figures 2(c)-(g) by comparing the black (observed) and grey (predicted) lines. In this case, the model has been trained on all data except these five profiles. The predicted (grey) profiles effectively capture the larger scale variations in the magnitude of κ_T, and often also successfully capture the small-scale spatial patterns.

Next, we use this ML model in combination with simulations to infer a data-driven estimate of diffusivity to be compared to the Richardson number based KPP diffusivity in the Drake Passage simulation. In this work, we use the internal wave-permitting model of the Drake Passage described in (Baker & Mashayek, 2022; Baker et al., 2023). The simulation is performed using the hydrostatic configuration of the Massachusetts Institute of Technology general circulation model (MITgcm, Marshall et al., 1997), with $0.01°$ horizontal resolution ($\sim 600 - 1100$ m) and 225 vertical levels with a variable resolution that is < 25 m at all depths less than 4500 m. The simulation is nested within a larger Southern Ocean simulation (Tulloch et al., 2014), from which it derives initial and boundary conditions, and is described fully and verified in Baker and Mashayek (2022); Baker et al. (2023).

The simulation outputs are, among others, temperature, salinity and the three components of the velocity as well as their gradients. Hence, a local value of the Richardson number can be computed from the simulation's output, a slice of which at $56.8°$ S is shown in figure 3(b). However, such a large-scale simulation, even though of relatively high resolution, does not resolve scales small enough to predict dissipation rates and associated turbulent diffusivities accurately. Therefore, the simulation uses the online KPP scheme to estimate a turbulent diffusivity κ_T (hereafter κ_{KPP}) with which scalar gradients are vertically diffused. Using the ML model, we obtain a second, offline, parameterisation of diffusivity (hereafter

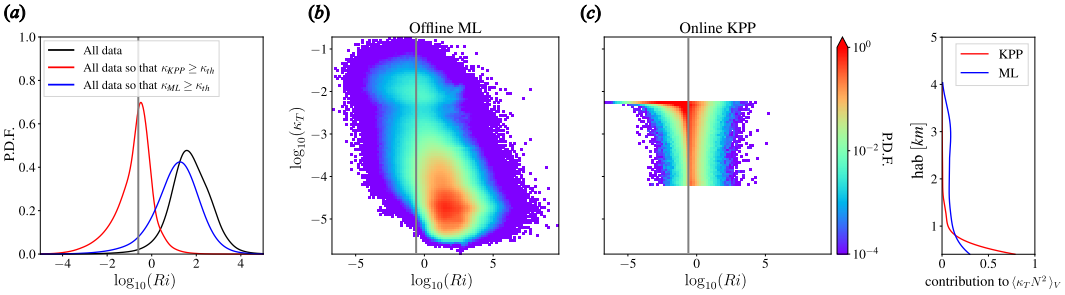


Figure 3. (*a*) Probability density function (PDF) of the Richardson number for the simulation of the Drake passage. In black, all data are shown, in red, only data in locations where κ_{KPP} is elevated above its background value of $\kappa_{th} := 5 \times 10^{-5}$ m^2 s^{-1}, and in blue, only data in locations where κ_{ML} is elevated above 5×10^{-5} m^2 s^{-1}. (*b*) Joint probability distribution of Ri and κ_{KPP}. (*c*) Same for Ri and κ_{ML}. (*d*) Contribution to $\langle \kappa_{\mathrm{T}} N^2 \rangle_V$ of a given height above the bottom (*hab*) for the KPP (red) and ML (blue) parameterisations. Note that the data is masked to exclude the top 500 m, since the KPP parameterisation involves a separate surface boundary layer component which is not of interest here.

κ_{ML}), which can be compared with κ_{KPP}. This comparison was previously made in Baker et al. (2023), where the water mass transformation in the simulation was calculated using both κ_{KPP} and κ_{ML}; further discussion of the validity of this parameterisation technique is provided therein.

Figure 2 shows profiles of diffusivity κ_{T} at one simulation time at the locations of the observational data — κ_{KPP} is shown in blue, and κ_{ML} in red. It is clear that κ_{KPP} gives a very crude estimate — throughout most of the water column, κ_{KPP} is constant at its background value of 5×10^{-5} m^2 s^{-1}. Elevated values of κ_{KPP} are found in the bottom few hundred metres, consistent with the lower right panel of figure 1. In comparison, κ_{ML} gives a diffusivity that varies more gradually over the water column, with magnitude much more similar to the observational profiles that are shown in black. Note that we do not expect (red) κ_{ML} to exactly match (black) observations, since diffusivity is temporally variable. We do notice that the small-scale variance in κ_{ML} is significantly less than both the small-scale variance of the (black) observations and the (grey) ML applied to observations. This suggests that, although the ML model can capture much of the small-scale variance in diffusivity, the simulation does not resolve all of the small-scale variance in the training scalar fields T and S. This is consistent with comparisons of this simulation with observations in Baker et al. (2023), and the consequences of this for parameterisation of κ_{T} will be discussed further later.

Using this framework, we are now well-equipped to compare the KPP and ML parameterisations in the Drake passage.

2.2 Results

Figure 3(*a*) shows a histogram of the Richardson number for the entire dataset (black). The mode is at $Ri = 27$. Restricting to only locations where κ_{KPP} is above its background value, i.e. KPP has 'turned on' shear or convective instability, gives the red curve, which has a mode just below the critical Richardson number Ri_{c} and is skewed towards small values of Ri. This is expected, since KPP elevates κ_{T} based on a critical Richardson number criterion. However, when this same restriction is placed on κ_{ML}, the distribution does not become skewed, although the mode approximately halves to 14, suggesting there does exist some lesser correspondence between high κ_{ML} values and Ri.

We present in figure 3(c,d) the joint probability distribution of κ_T and Ri for the KPP and ML outputs. As expected from the definition of κ_KPP, and as per the marginal instability paradigm, the KPP output clusters around the critical value of the Richardson number Ri_c. In comparison, the ML output is richer. The great bulk of data points seems to cluster in the low κ_T, high-Ri region (i.e. for Richardson number flows that are in theory statically stable). Whilst the KPP diffusivities in figure 3(c) (by definition) can be separated into 'high κ_T' and 'low κ_T' regimes by Ri_c, clearly this is not possible for the ML diffusivities, even by varying the value of Ri_c. This suggests that, if the ML diffusivity is to be believed, a marginal-stability based paradigm of mixing may not hold in this flow regime, at the scales resolved by the simulation.

Where are these differences coming from? To answer this question, we plot in figure 1 a vertical slice of the KPP and ML outputs. The KPP output seems to recognise shear and convective instability, and therefore 'turns on' κ_T near the bottom topography, but is 'blind' in the mid- and upper-ocean. On the contrary, the ML output seems to capture various interesting processes both near the bottom and in the mid- and upper-ocean. A relevant question to ask is: is the contribution to the bulk value of the vertical buoyancy flux $\kappa_\mathrm{T}N^2$ (proportional to ϵ via equation 1, denoted $\langle\kappa_\mathrm{T}N^2\rangle_V := \sum_V \kappa_\mathrm{T}N^2$ in the subsequent, with V corresponding to all the points in the simulation) from the processes 'unseen' by KPP important or negligible? Figure 3(d) shows that, while the contribution to $\langle\kappa_\mathrm{T}N^2\rangle_V$ comes mainly from the region close to the topography in KPP (i.e. for $hab \leq 2000$m), contributions are spread throughout the water-column in ML. Although some of the 'missing' mixing in the KPP parameterisation is likely due to unresolved small-scale shear in the simulation, which would be captured at higher resolution, the ML parameterisation shows that resolving these small-scale processes is not necessary to obtain a reasonable estimate of diffusivity.

All in all, whereas a Richardson-based approach captures some shear-driven mixing near the bottom topography (where the stratification is quite weak compared to the rest of the water column, and shear is high due to flow-topography interactions), it seems to be inaccurate in the rest of the water column, at least when applied at the resolution scale of the simulation. This raises the question of how applicable an inevitably small-scale Richardson number approach is to large-scale, low-resolution models. Indeed, although it is impossible to rule out the importance of small-scale, unresolved, shear in the simulation, increasing the resolution of climate models is not an appropriate answer yet, for practical reasons. Similarly, a Richardson number approach potentially neglects other processes that might be just as important (in setting the statistics of dissipation rates) as shear-driven instabilities in the mid- and upper-ocean, especially when all of these processes are not resolved, as is the case in many practical applications of such Ri-based parameterisations.

3 Discussion and outlook

In Mashayek, Baker, et al. (2022) the authors argued that the deep ocean circulation restratifies mixing hot spots at just the right rate to keep the flow marginally stable to turbulent mixing. The relatively high resolution of the model used in that work was key to showing the feedback between mixing and restratification needed to achieve such balance. Here, we add an important caveat: such a balance existed due to the specific stability-based nature of the mixing parameterization (simple Ri-based KPP) deployed in the model. Using the same model output but with the addition of a process-agnostic mixing parameterization (based on a machine learning model 'fitted' to global microstructure data), we show that conventional marginal and stability-based parameterisations of ocean mixing might be prone to spatial biases about where turbulent mixing is generated within the water column. At the scales considered in this study, such approaches seem to capture the overall turbulent statistics near the bottom boundary but are inaccurate in the rest of the water column. More precisely, while it is not possible to rule out the importance of small-scale shear-induced turbulence that is not resolved by current large-scale ocean models, our study suggests that at the scales resolved by ocean models, current parameterisations of mixing based solely

on marginally unstable shear-instability might be biased towards weakly stratified regions. Whereas they might hold at small scales, our study suggests that these parameterisations do not yet accurately represent on large scales the small-scale effects that they strive to describe, especially when the small scales are not accessible. This emphasises the need to understand better how small-scale turbulence translates into large-scale parameterisations.

This also suggests that other physical processes might be as important as shear instabilities in setting turbulent statistics in the water column. We speculate such contributions might originate from double-diffusive type instabilities (i.e. double-diffusive convection and salt-fingering). For example, Middleton et al. (2021) developed a method for estimating ϵ from temperature and salinity measurements in the double-diffusive convection regime. Similarly, whereas shear-driven turbulence seems to adjust towards a marginal Richardson number $Ri \sim 1/4$ during a short phase of its life-cycle (namely the 'Goldilocks' phase (Mashayek et al., 2021)), it can still be energetic in its fossilised phase, for which the flow has essentially restratified. Time-variation of shear (due to e.g. internal waves in the ocean) can also have an impact on the turbulence energetics and mixing induced by a shear-driven instability, as numerically shown by Lewin and Caulfield (2022). More precisely, the authors found that for a decelerating shear, mixing can still be very efficient even though the flow has $Ri > 1/4$ (and in fact more efficient than in the $Ri \lesssim 1/4$ phase of a shear-instability with fixed, time-independant, shear).

Note that we have focused here on potential biases arising in the prediction of turbulent diffusivities κ_T when inferred from measures of *momentum* dissipation (such as ϵ or ν_T) rather than actual measures of *scalar* mixing (such as χ), through the commonly used $\Gamma \simeq 0.2$ assumption. This assumption might introduce other biases. For instance, by considering direct numerical simulations, Couchman et al. (2023) and Petropoulos et al. (2024) showed that focusing on the relatively weakly stratified regions of the flow that spontaneously develop in stably stratified turbulent flows might well significantly underestimate the statistics of χ (quantifying scalar mixing) because of the presence of strongly stratified interfaces where mixing can actually be both intense and 'efficient' in the sense that the flux coefficient $\Gamma \gg 0.2$. This point is further exacerbated by the fact that such strong density gradients in the vicinity of interfaces might well be characterised by $Ri > 1/4$, effectively making these regions of interest invisible to current mixing analysis and parameterisations.

All in all, our study emphasizes the need to understand better and characterise more carefully the different small-scale processes contributing to the turbulent and mixing statistics in the ocean interior and their impact on larger scale statistics. A bi-product of our work concerns the applicability of a ML-based parameterization to outputs of relatively high-resolution ocean models. While microstructure profilers show continuous turbulence signals, static-stability-based criteria for mixing (such as KPP-like parameterizations) only produce mixing when there is significant shear. The ML-based parameterisation considered here is able to recover the 'continuum' of turbulence throughout the water column and closely mimics observations in the Southern Ocean. This work sheds light on the reasons behind the shortcomings of Richardson number-based parameterisations and alludes to the utility of process-agnostic machine-learning approaches to infer mixing from the wealth of observations of density fine scales provided by global observational programs such as Argo and GO-SHIP. This can 'buy' us time while we further our understanding of isolated and combined contributions of individual processes to mixing and inform such understanding by identifying regions of potential mixing without the need to believe in the values produced by the ML-based parameterization.

Open Research Section

The simulation data used in this work was based on Baker et al. (2023) and can be found at `https://doi.org/10.1029/2022AV000858`. The MITgcm is documented at `https://`

`mitgcm.readthedocs.io` and available at `https://doi.org/10.5281/zenodo.1409237`. The BBTRE microstructure data can be obtained from `https://microstructure.ucsd.edu/`. No new data was produced for the purpose of this work.

Acknowledgments

This project received funding from the European Union's Horizon 2020 research and innovation program under the Marie Sklodowska-Curie Grant Agreement No. 956457. LB would like to acknowledge funding from EPSRC grant EP/X028135/1. AM would like to acknowledge funding from NERC grant NE/P018319/1 and ONR grant N00014-22-1-2082.

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